

MINI-PROJECT 2 SHORT TECHNICAL REPORT

Course: Cross-Platform Mobile App Development (VKU)
Title: Real-time Study Room Booking App (React Native & Expo)
Weeks: 5-6 Weight: 10%
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1. GENERAL INFORMATION & DELIVERABLE LINKS

Solo project. Nguyễn Văn Duy (23IT038) built the Expo app, booking rules, Cloudflare KV API, and this report. Contribution: 100%.

Live Demo URL: <https://vku-room-booking-part2.pages.dev/>

API health: <https://vku-room-booking-part2.pages.dev/api/health>

GitHub: <https://github.com/minhduy6868/mob-lab-room-booking>

Video: Not recorded. The live URL is the demo deliverable.

Demo login is pre-filled: abc@vku.udn.vn / vku@2026 (Nguyễn Văn Duy, 23IT038). Week 5 remains at <https://vku-room-booking.pages.dev/>. Week 6 is the part2 host above.

2. FEATURE IMPLEMENTATION CHECKLIST

#	Required feature	Status	Where it lives
1	Room feed: photo, building/floor, capacity, Available vs Occupied	Complete	FlatList in BrowseRoomsScreen. RoomCard is memoized. windowSize 5, initialNumToRender 10. Status comes from KV bookings for the selected day.
2	Search and chips: building A/B/C/V, capacity 2-20, equipment	Complete	Search matches name, building, amenity, description. Chips: type, building (A, B, C, V, library, dorm), capacity 2-20 or over 20, projector / whiteboard / high-spec PC / AC, available vs full.
3	7-day selector and 2-hour slots with conflicts disabled	Complete	DateStrip shows 7 days. Slots: 07:30-09:30, 09:45-11:45, 13:00-15:00, 15:15-17:15, 17:30-19:30, 19:45-21:45. Occupied, past, overlap, daily limit (2), and maintenance slots are disabled. Rules run in booking-rules.ts and functions/ lib/rules.js.
4	Booking pass with QR check-in	Complete	POST /api/bookings returns a booking code. BookingConfirmation is a modal pass. QRCodePassModal opens from My Bookings. Check-in is allowed from 10 minutes before the slot.
5	Zustand: session, reservations, filters, cancel	Complete	useAuthStore holds the session. useBookingStore holds bookings, date, filters, book, cancel, and check-in. TanStack Query holds the room catalog and the live booking cache.
6	Persist with AsyncStorage	Complete	Zustand persist uses @react-native-async-storage/async-storage (localStorage on web). Auth session and booking/filter cache are restored on the next launch. KV remains the source of truth.
7	Check-in reminder 15 minutes before the slot	Complete on device	expo-notifications schedules a local alert 15 minutes before start and cancels it if the booking is cancelled. The Pages web demo does not show an OS notification; Expo Go / a dev build does.
8	Navigation, motion, list performance	Complete	Root stack + bottom tabs. Typed params for RoomDetails and BookingConfirmation. Reanimated FadeInDown on cards. Swipe-left cancel uses Gesture Handler.

3. TECHNICAL ARCHITECTURE

One Expo SDK 57 TypeScript app. Web export is hosted on Cloudflare Pages. The same screens run in Expo Go. A booking is accepted only after Pages Functions write Cloudflare KV. The client then upserts that record into Zustand and the TanStack Query cache, so the open session updates from the response. Another browser sees the change on focus, reconnect, or pull-to-refresh. There is no 2.5 second poll and no WebSocket.

App.tsx Navigation, Query, alerts; notification handler on iOS/Android
src/navigation/ Root stack + tabs; RoomDetails; BookingConfirmation modal
src/screens/ Login, Browse, RoomDetails, MyBookings, Profile
src/store/useAuthStore.ts Session
src/store/useBookingStore.ts Filters, book, cancel, check-in, reminder hook
src/lib/booking-rules.ts Conflict engine (client)
src/lib/reminders.ts 15-minute local notification
src/lib/storage.ts AsyncStorage adapter
functions/api/bookings.js GET, POST, PUT cancel|checkin
functions/_lib/rules.js Same conflict rules on the server
wrangler.toml KV binding VKU_BOOKINGS

Flow: login -> hydrate bookings from GET /api/bookings -> pick a day and a free slot -> POST /api/bookings. The server returns 200 plus a booking code, or 409 plus a Vietnamese conflict message. Cancel is PUT with action cancel, and is refused inside 30 minutes of the start time.

4. SCREENSHOTS

Captured from <https://vku-room-booking-part2.pages.dev/> at 390x844 on 30/09/2026. Login used the pre-filled demo account.

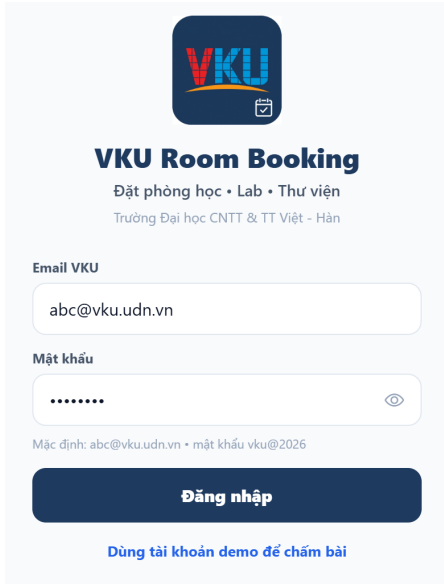


Figure 1. Login. Email abc@vku.udn.vn and password are pre-filled.

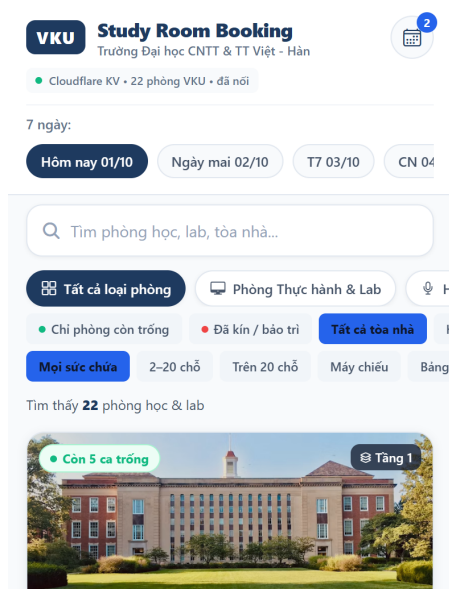


Figure 2. Browse. 7-day strip, building/capacity/equipment cards.



Figure 3. Room details. Slot list disables occupied and past slots.



Figure 4. My Bookings. Pull to refresh. Swipe left cancels a valid booking.

5. NOTES

GitHub origin/main is still the Week 5 commit (56ed666). The Week 6 app is what is deployed on the part2 URL. Push the local tree before the repository itself is graded.

KV list() is briefly delayed after a write, so a second device can miss a booking for a few seconds until focus or pull-to-refresh. The web host cannot fire expo-notifications; the reminder code runs on iOS and Android.